

```
#include <stdlib.h>

#define MAX 5

int stack[MAX];

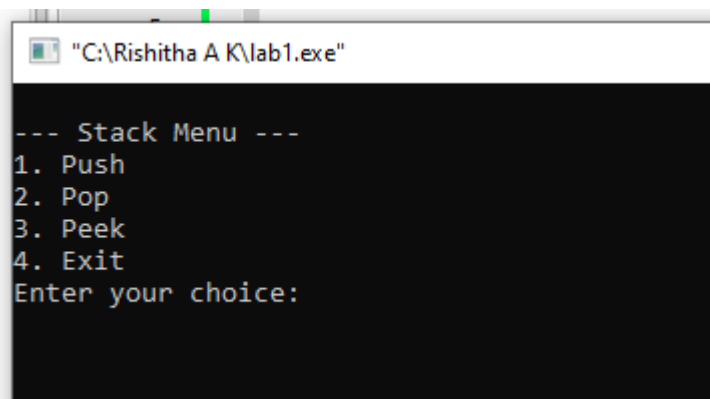
int top = -1;

void push(int value) {
    if (top == MAX - 1) {
        printf("Stack Overflow! Cannot push %d\n", value);
    } else {
        top++;
        stack[top] = value;
        printf("%d pushed onto the stack.\n", value);
    }
}

void pop() {
    if (top == -1) {
        printf("Stack Underflow! No elements to pop.\n");
    } else {
        printf("Popped element: %d\n", stack[top]);
        top--;
    }
}

void peek() {
    if (top == -1) {
        printf("Stack is empty.\n");
    } else {
        printf("Stack elements (top to bottom):\n");
        for (int i = top; i >= 0; i--) {
            printf("%d\n", stack[i]);
        }
    }
}
```

```
int main() {  
    int choice, value;  
    while (1) {  
        printf("\n--- Stack Menu ---\n");  
        printf("1. Push\n");  
        printf("2. Pop\n");  
        printf("3. Peek\n");  
        printf("4. Exit\n");  
        printf("Enter your choice: ");  
        scanf("%d", &choice);  
        switch (choice) {  
            case 1:  
                printf("Enter value to push: ");  
                scanf("%d", &value);  
                push(value);  
                break;  
            case 2:  
                pop();  
                break;  
            case 3:  
                peek();  
                break;  
            case 4:  
                printf("Exiting program.\n");  
                exit(0);  
            default:  
                printf("Invalid choice! Please try again.\n");  
        }  
    }  
    return 0;  
}
```



```
"C:\Rishitha A K\lab1.exe"

--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Exit
Enter your choice:
```